

Augmenting Office Workers with Robot Avatars - An Exploratory Study

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Abstract

This paper explores personalization of telepresence robots as an exploratory step towards physical representations of augmented humans in remote collaboration. We investigate how individuals design and use robot avatars that express identity and presence through clothes and accessories. Our participatory design study involved eight participants from the same work community who each used a personalized humanoid robot avatar in an office setting for one day. The findings reveal that video-based face visibility enhanced recognizability, while personalized clothing made the robot seem more human and approachable. Participants customized their avatars to reflect personal style, align with work context, or appear friendly.

CCS Concepts

• **Human-centered computing** → **User studies; Empirical studies in HCI**; *Collaborative and social computing design and evaluation methods*; • **Computer systems organization** → *Robotics*.

Keywords

Robots, avatar, user experience, telepresence, participatory design, augmented humans, office

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1 Introduction

As human-machine integration advances, robot avatars offer a tangible entry point toward physical augmentation in remote collaboration. These avatars act as embodied extensions of individuals, enabling presence, movement, and identity expression beyond the

body. In this paper, we consider personalized telepresence robots as a first step toward augmented humans—systems that blend physical embodiment with social and cognitive continuity across distance. Telepresence robots are already used in domains, such as medicine [19], education [25], and academic conferences [22]. The COVID-19 pandemic accelerated the adoption of remote working tools [1], prompting organizations to rethink long-term models for remote and hybrid work.

We explore the concept of personalized robot avatars and how they can support remote work through expressive and identity-linked embodiments. Existing telepresence designs typically feature a generic screen-on-wheels or humanoid form, with fixed appearance regardless of the user [27]. While social robots are often defined as autonomous systems that follow human social norms [3], telepresence robots are remote-controlled and can instead be seen as extensions of their operators. Kristoffersson et al. [15] describe Mobile Robotic Telepresence as enabling a remote person to navigate and interact in a shared space. We use the term robot avatar to describe a physical proxy that embodies and communicates on behalf of its remote user.

This paper presents a participatory design study with eight participants from a shared work community. Each participant co-designed a personalized robot avatar and used it in their office for one day. Our contribution includes:

- the first in-the-wild participatory design study on personalizing telepresence robots in an office context,
- insights into clothing and accessory preferences and the motivations behind them, and
- findings that personalization improved social presence, recognition, and perceived humanity of the robot avatars.

Our research contributes a more personal and customizable design approach, giving remote users a way to reflect their style, social cues, or work context through the robot.

2 Related Work

The use of avatars to extend presence and identity has long been established in digital settings, and is increasingly being explored through physical embodiments. Personalized telepresence robots represent an early step toward systems that physically augment the user, reflecting their appearance, style, and social presence in shared environments.



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2.1 Telepresence Robots and Social Presence

Telepresence robots have been studied in diverse contexts including healthcare [19], education [25], and conferences [22]. In office settings, they support informal communication, though recognition of the remote user often requires video presence [16]. The telepresence experience has been investigated both from the viewpoint of people surrounding the robot, as well as of the operator, as a telexistence providing a realistic sensation of existence in remote places [26]. Muratbekova-Touron and Leon [20] observed that telepresence robots are often anthropomorphized by both operators and colleagues, suggesting a shared identification process. Groom et al. [10] emphasized that operators prefer robots they have designed themselves, underlining the significance of personalization for user embodiment.

Technological extensions such as VR or wearable-controlled avatars further blur the line between telepresence and embodied augmentation [7, 12]. These studies illustrate a trajectory from telepresence toward augmenting human capabilities via remote-controlled robotic systems.

2.2 Digital Avatars and Identity Expression

In virtual environments, users customize avatars to express identity or explore alternative selves [8, 13, 17]. While digital avatars often deviate from realism, their design is influenced by social norms and context [17]. Neustaedter and Fedorovskaya [21] identified different identity strategies in avatar creation, and McArthur [18] called for more UX-focused research on customization tools. These findings suggest that the principles behind virtual avatar expression may inform physical avatar design, but are likely to manifest differently when embodiment involves real-world presence and recognition. On the other hand, when comparing different digital avatar designs in VR and AR, it has been found that a photorealistic full-body human avatar of the other person was preferred by the user [24].

2.3 Robot Avatars and Personalization

Robot avatars have occasionally appeared in entertainment and science fiction, but research on physically personalized telepresence systems remains limited. Prior work has examined robot-mediated gestures [6] and personality projection [5]. Neustaedter et al. [22] briefly explored appearance customization, while others have demonstrated personalized clothing in workplace contexts [14] and performative settings [23].

Fitter et al. [9] showed that personalization is not universally welcomed, as non-operators sometimes reacted negatively to features like wigs. Their study highlighted the importance of context and cultural sensitivity. Unlike Fitter et al., our work involved participants actively dressing a humanoid robot with their own clothes and using it in a shared office over a normal workday. Our participants were embedded in a work community, and recognition based on clothing alone was common.

Taken together, prior studies lay the groundwork for understanding robot avatars as extensions of the self. Our work builds on these efforts by showing how clothing and appearance customization contribute to social presence in everyday work settings, and by framing this as a move toward physically embodied systems that may eventually augment human functions, not just represent them.

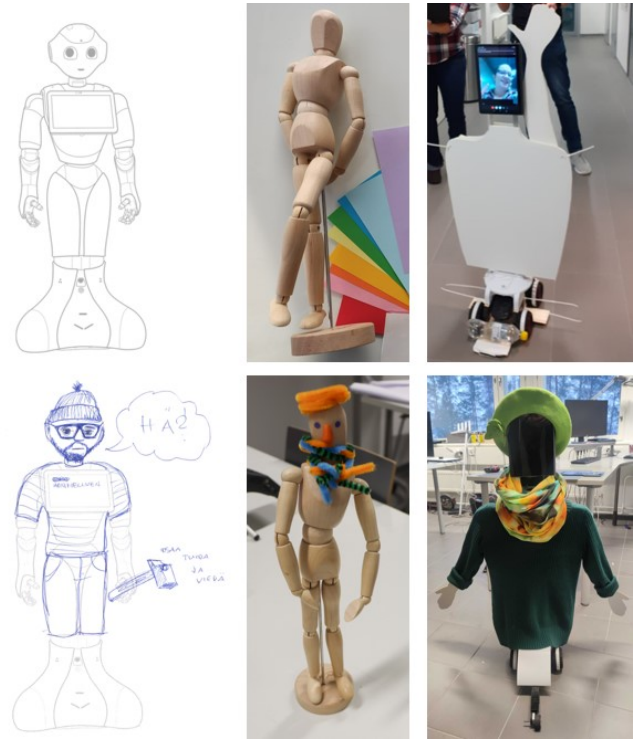


Figure 1: PD methods and example outputs: sketching, low-fi prototyping, and robot dressing.

3 Research Method and Set-up

3.1 Study Overview and Participants

The study involved eight participants from the same work community, who each personalized and operated a telepresence robot in their office setting. The process began with a focus group to gather early design ideas, followed by individual participatory design (PD) sessions using sketching and low-fi prototyping. A humanoid robot avatar frame was built to support physical customization with clothes and accessories. Each participant used the personalized robot for one office workday, after which post-study interviews were conducted. The robot frame created for the study was mounted on an RC car chassis with a 3D-printed mount for a tablet used in video calls. The structure supported wearable items like scarves and hats (Figure 1).

The participants ($n=8$; 3 women, 5 men) were aged between 26 and 55+, with most in the 36–45 range. All worked in the same corridor and department. Most had prior experience with RC devices, and several had used digital avatars in online platforms. Remote work was common, typically once per week.

3.2 Study Tasks

We adopted a participatory design approach [4] to explore user-driven personalization preferences. The PD actions included (Figure 1):

- Sketching robot avatars using a self-expression template [2],
- Low-fi prototyping with mannequin dolls and craft materials,

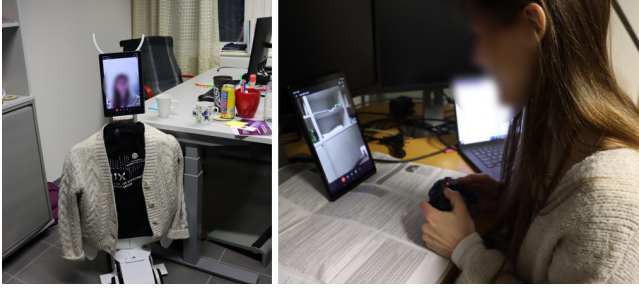


Figure 2: Using the personalized robot avatar in a shared office environment.

Table 1: Clothes and accessories used for personalization by each participant (n=8).

No.	Robot Personalization	No.	Robot Personalization
1	t-shirt, wool jacket	5	shirt
2	jumper, scarf, hat	6	jacket and scarf
3	scarf, hat, candy box	7	hat and bowtie
4	vest	8	t-shirt and hat

- Dressing a physical robot prototype (done at the start of the office study day).

The office study day was agreed with each participant. During the office study, participants remotely operated their dressed-up avatar robot for 2–4 hours. They navigated the space, interacted with coworkers, and held at least one conversation using the robot (Figure 2). Afterward, they completed a questionnaire and participated in a semi-structured interview, reflecting on both their own and others' avatar designs.

4 Findings

4.1 Personalizing the Robot Avatar

Participants focused their customization efforts on the upper body — especially the head, neck, and face areas — during sketching and low-fi prototyping (Figure 1). Some drew the robot holding items or added features such as hair (2/8), glasses (2/8), hats (3/8), and scarves (3/8).

For the physical robot, participants brought clothing and accessories from home (Table 1) and dressed the robot prior to its use in the office. The personalization clustered around the torso and head, and hats (4/8) and scarves (3/8) were especially common.

Participants explained their clothing choices as reflecting personal style (P2, P5), work context (P8), or a desire to make the robot appear more friendly and less robotic (P1, P3): *"It looks cute and less scary" (P3); "brings color and personality to it" (P1).*

4.2 Experiences at the Office

Initial reactions varied. Some participants found the experience natural (P1, P4, P7), while others described it as alien (P5) or slightly intrusive (P6). One participant noted positive attention: *"People were curious and interested" (P2).*



Figure 3: Coffee room interaction using the robot avatar.

The main challenge was the robot's mobility. Seven participants reported difficulty with turning, doorways, or video call delays. Suggestions included adding a rotatable head or the ability to knock on doors. One noted cognitive strain from needing constant camera awareness (P8).

Despite limitations, many appreciated the social connection enabled by the robot. Participants noted faster interactions than online chat (P5), being noticed at the office (P1, P6), and the novelty of dual presence (P4): *"You can be in two places at the same time".*

4.3 Recognizing Others and Interaction Quality

When shown photos of others' avatars, most participants recognized at least one colleague by their robot's style or clothing. Personal items and logos aided recognition (P3), though some noted that hats and outerwear felt out of place indoors (P4, P8).

Interactions were described as cheerful and engaging. Participants reported receiving smiles and compliments (P4, P7), and one stated: *"[It] brought added value. Gave a new dimension to remote work" (P7).* Personalization was consistently linked to increased social presence: *"Clothes made it feel more human" (P8).*

5 Discussion

5.1 Humanizing Telepresence Robots

Participants consistently reported that personalization made the robot avatars more relatable and improved social interaction. Dressing the robot introduced warmth and personality to what might otherwise be perceived as a cold interface. Several participants noted that being present via a personalized avatar felt more engaging than attending remotely via conventional video tools. Participants often dressed the robot avatars to match their personal or professional identity. This included everyday clothing, workplace-appropriate garments, and accessories that made the robot appear friendlier. Personalization clustered around the torso, head, and neck regions that are most visible in face-to-face interaction.

While prior research by Fitter et al. [9] found that some personalization (e.g., wigs) could provoke negative reactions, our study avoided such extremes. Participants gravitated toward familiar, neutral styles, which likely contributed to the consistently positive feedback. Showing the user's face on the tablet also helped reduce

ambiguity and made the robot easier to recognize, potentially reducing any uncanny effect. Customizing digital avatars is common in games and social media, often reflecting aspirational or stylized versions of the self [8, 13]. In contrast, our participants tended to create robot avatars that closely matched their everyday appearance. This shift toward realism may reflect the avatar's physical co-presence and social accountability in shared workspaces.

Our research setting differed from large-scale deployments like Neustaedt et al.'s conference study [22]. Because participants shared a physical office culture, recognition and interaction followed their usual social patterns. In our study the robot temporarily occupied the user's workspace, reinforcing the impression of presence.

5.2 From Robot Avatars to Augmented Humans

As robot avatars evolve to include movement, sensing, and interaction capabilities, they may come to function as physical augmentations of the user, not just representations. This aligns with recent calls to see robots and humans as interdependent agents in shared activity systems [11]. It also raises questions about social expectations, etiquette, and identity in human–robot collaboration. Cultural and contextual factors will likely shape how far personalization can or should go in order to create an augmented human experience. As remote interaction technologies become more embodied, the boundary between representation and augmentation will blur.

Even within the constraints of our low-tech robot frame, participants showed enthusiasm and creativity. The sketching and prototyping sessions revealed that participants valued expressivity and envisioned expanded functions (e.g., moving arms, object handling). The expressivity could also be enhanced by integrating physiological sensor data to adapt the behavior and appearance of the robot avatar. This would strengthen the link between the operator and the remote agent, and potentially enable richer vocabulary and stronger perception of authenticity for the embodied telepresence.

5.3 Limitations

This study involved a small participant group in a single work community, and each user interacted with the robot for only one day. A larger sample and longer deployment would help validate the findings. In addition, our prototype had no articulated limbs or legs, limiting both personalization options and interactive capabilities. Despite these limitations, this is—to our knowledge—the first study to explore in-the-wild personalization of telepresence robots with clothing and accessories in a work setting. The findings can inform future exploration of personalized embodiment in telepresence and physical avatar systems.

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